

A Framework for the Integration of User Centered Design and Agile Software Development Processes

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ABSTRACT

Agile and user centered design integration (AUCDI) is of significant interest to researchers who want to achieve synergy and eliminate limitations of each. Currently, there are no clear principles or guidelines for practitioners to achieve successful integration. In addition, substantial differences exist between agile and UCD approaches which pose challenges to integration attempts. As a result, practitioners developed individual integration strategies. However, success evaluation of current AUCDI attempts has been anecdotal. Moreover, AUCDI attempts cannot be generalized to provide guidance and assistance to other projects or organizations with different needs.

My thesis aims to provide a Software Process Improvement (SPI) framework for AUCDI by providing generic guidelines and practices for organizations aspiring to achieve AUCDI in order to address AUCDI challenges including: introducing systematicity and structure into AUCDI, assessing AUCDI processes, and accommodating project and organizational characteristics.

Categories and Subject Descriptors

D. Software, D.2 Software Engineering, D.2.1 Requirements Specification, D.2.10 Design, H.5.2 User Interfaces []

General Terms

Design, Human Factors

Keywords

Agile, User Centered Design, Framework

1. INTRODUCTION

Agile processes have evolved to address some of the perceived limitations of plan-driven approaches. Agile processes are iterative and incremental. None of the existing agile processes explicitly include principles and practices for understanding and eliciting testable and verifiable usability and user experience requirements. Similarly, design practices for

satisfying requirements for usability and user experience or practices for evaluating agile systems for usability and user experience are generally absent. In general, it is not clear yet how to incorporate UCD into agile processes without sacrificing the acknowledged benefits of each of these individual processes.

UCD approaches recommend a number of principles of design, including early focus on users and tasks, empirical measurement and iterative design. UCD approaches enable developers to understand the potential users' needs of their software, and how their goals and activities can be best supported by the software.

The importance of incorporating such an approach into software development lifecycles has led to research investigating possible approaches on how to integrate agile and UCD processes. Nevertheless, the review of AUCDI research [1-4] suggests that there is currently an absence of both a formalization of what constitutes AUCDI, and clear and explicit guidelines that define what comprises successful AUCDI. The research in my thesis aims to provide an SPI framework for achieving AUCDI.

The rest of this paper is structured as follows: section 2 will present a number of AUCDI research questions. Section 3 highlights our research hypothesis and provides an overview of the framework including goals, expected contribution and components, and section 4 discusses the evaluation plans.

2. AUCDI THEMATIC QUESTIONS

Based on a substantial domain analysis of literature, focusing on previous attempts at AUCDI, we identified the following research questions raised by researchers addressing AUCDI.

- When and how to conduct user research in agile projects in a manner that fits the tight agile timelines?
- How to design user interaction in agile projects?
- How to evaluate usability and user experience of software developed in agile projects?

As a result of our analysis, we have identified a set of thematic questions not addressed directly in the literature, whose answer can play a significant role in better formalizing and structuring AUCDI efforts. These questions can be separated into two groups: those focusing on accommodating project and organizational characteristics; and those on measuring and assessing AUCDI efforts.

In the former category, there are significant questions related to how to determine if an organization is sufficiently

mature for AUCDI in a specific project. There are also questions associated with how to identify the most appropriate AUCDI practices for adoption, and the potential difficulties that could develop during adoption.

In the second category (measurement and assessment), there are important questions to be answered related to how to judge and measure the success of an AUCDI process, as well as on how to identify improvement areas related to AUCDI.

3. FRAMEWORK OVERVIEW

Our research hypothesis states that "High user centered design maturity level is directly proportional with high AUCDI maturity levels". In order for us to test this hypothesis we proposed an AUCDI framework.

Many sources have contributed to formulating this framework, including a synthesis of key tensions between agile and UCD processes and previous publications on AUCDI attempts [1-4]. In addition to a set of empirical data reached from in-depth, semi structured interviews with participants from 11 companies that attempted AUCDI.

3.1 Framework Goals and Expected Contribution

The AUCDI framework has a number of goals and expected contributions which are summarized as follows

- Introduce formalization and structure to AUCDI efforts.
- Accommodate project and organizational characteristics influencing AUCDI adoption efforts.
- Measure and assess AUCDI.
- Help organizations that have adopted AUCDI to improve their AUCDI process maturity levels.

3.2 Framework Components

The framework consists of two components; the AUCDI capability maturity model, and AUCDI framework implementer. More details about those components will be provided below

3.2.1 The AUCDI Capability Maturity Model (AUCDI - CMM)

This component serves to measure and assess the organization's AUCDI potential, provide a scale for identifying the target AUCDI maturity level for a project aspiring to adopt AUCDI and provide a current state analysis to evaluate organization's AUCDI strengths and weaknesses. AUCDI-CMM includes 5 components: a reference model, assessment process, performance scale, assessment guidelines and level improvement guidance.

The first component is the AUCDI reference model, which is used to define the evaluation criteria that determine AUCDI maturity levels and thus need to be included/ examined in the AUCDI assessment. This component will also identify impediments to successful adoption of AUCDI practices in order to eliminate them before continuing with the AUCDI effort.

The second component is the AUCDI assessment process which is used to conduct a current state analysis to identify the strengths and weaknesses of both an organization and project related to AUCDI, and thus provide a basis for planning and implementing improvement action.

The third component is the AUCDI performance scale that is used to rate organizational AUCDI performance elements. The closer the organization meets the requirements of the AUCDI reference model, the higher its ratings.

The fourth component is the AUCDI assessment guidelines which provide guidance to carry out AUCDI assessment, interpret the findings, and generate and present results.

The final component is the AUCDI level improvement guidance which uses the output of previous phases to provide clear guidance for achieving higher AUCDI maturity levels.

3.2.2 AUCDI Framework Implementer

This individual will be responsible for applying the proposed framework and should have sufficient knowledge in both agile and UCD methods.

4. PLANNED EVALUATION

The framework will be validated in an industrial context to evaluate its feasibility, effectiveness, advantages and disadvantages, and the results will be used to improve it.

In the preliminary evaluation phase surveys will be sent to agile and UCD practitioners working on projects that require AUCDI. This survey results will help validate and improve the framework.

In secondary evaluation phase the framework will be validated by a number of organizations aspiring to achieve AUCDI through utilizing the framework in one of their upcoming projects. The framework will be used to measure and assess the AUCDI potential of the organization and to provide a scale for identifying the target AUCDI level for the project aspiring to adopt AUCDI. In addition, the framework will help identify organizational impediments to successful adoption of AUCDI practices in order to eliminate them before continuing with the AUCDI adoption effort. The secondary evaluation phase will be conducted via in-depth, one-to-one semi structured interviews involving participants who have utilized our framework to achieve AUCDI.

5. REFERENCES

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